

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – 4.2 PROBABILITY I: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How is probability applied in real life situations? You have worked with the KMD forecast of a 70% chance of rain in Kericho and the Rift Valley on a March morning before the long rains. Wanjiru Chebet (a maize farmer near Eldoret with 2 acres), Captain Otieno (a pilot at JKIA flying to Kisumu) and Ms Achieng (Nairobi County disaster office) must each decide what to do. Write a Final Explanation in your own words using Claim, Evidence and Reasoning. Use the numbers and methods from Lessons 1 to 9. Show your working for every calculation and write probabilities as fractions or decimals between 0 and 1. Answer each section in the space under its prompt.

Section 1: What does a 70% forecast mean?

The KMD forecasts a 70% chance of rain. (a) Does this guarantee rain on that day? Give a claim. (b) Use the Kericho March record (rain on 14 days out of 20) to show how a number like 0.7 can be worked out. (c) In the coin activity, the running relative frequency of heads was 0.6 after 10 tosses, 0.55 after 20, 0.55 after 40 and 0.525 after 80 tosses. Explain what this pattern tells you about trusting estimates from long records.

Write your answer here:

Section 2: Sample space, events and theoretical probability

(a) List the sample space S for the weather on one March day in Kericho, using the three day states from class. (b) Write the event 'rain, as the farmer sees it' as a subset of S . (c) For a fair coin and for a fair die, calculate $P(H)$ and $P(\text{even number})$ using theoretical probability. (d) Our class got 42 heads in 80 tosses (0.525) but theory gives 0.5. Explain why they differ. (e) Can we find $P(\text{rain})$ for the weather by counting outcomes in S in the same way as for the die? Explain.

Write your answer here:

Section 3: Combining probabilities on one day

(a) Is 'rain' and 'dry' on the same day at the same place mutually exclusive? Give a reason. **(b)** Use $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$ to find $P(\text{rain only or storm})$ and $P(\text{storm or dry})$. Name the law you used and say why it is allowed. **(c)** Find $P(\text{no rain})$ using the complement rule. **(d)** How does your answer to (b) agree with the KMD forecast of 70%?

Write your answer here:

Section 4: Two days of weather

Wanjiru wants to know about the next two days. Assume the days are independent, with $P(R) = 0.7$ and $P(N) = 0.3$ on each day. (a) Why is the multiplication law suitable here? (b) Draw a probability tree diagram for two days and write the probability of each of the four outcomes. Check that they add to 1. (c) Find $P(\text{exactly one rainy day})$ and $P(\text{at least one rainy day})$. Show a second way of getting $P(\text{at least one rainy day})$. (d) Why is $P(\text{rain on both days})$ smaller than $P(\text{rain on one day})$?

Write your answer here:

Section 5: Final Explanation – advising the decision makers

Using the class decision rules, advise each person. Farmer Wanjiru plants if $P(\text{rain on each of the next two days}) \geq 0.49$ is considered good, or if $P(\text{at least one rainy day in two}) \geq 0.9$. Captain Otieno delays or reroutes if $P(\text{storm}) \geq 0.1$. Ms Achieng prepares drainage and warns residents if $P(\text{rain}) \geq 0.7$ or $P(\text{storm}) \geq 0.1$. (a) State what each person should do and give the probability evidence. (b) Write a Claim, Evidence and Reasoning answer to the driving question: How is probability applied in real life situations? In your reasoning explain how the numbers are worked out and combined, and why people can still act sensibly if the day turns out differently.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Meaning of probability and relative frequency	Clearly explains that 70% is a long-run likelihood, not a guarantee, and uses $14/20 = 0.7$ and the running relative frequencies correctly to show that more trials give steadier estimates.	States that the forecast is not a guarantee and calculates $14/20 = 0.7$ correctly, but gives limited explanation of why more trials matter.	Treats 70% as a guarantee or does not link it to past records or relative frequency.
Sample space, events and theoretical probability	Lists $S = \{\text{rain, storm, dry}\}$, writes events as subsets, calculates $P(H)$ and $P(\text{even})$ correctly, and explains both chance variation and why the weather states are not equally likely.	Lists the sample space and calculates theoretical probabilities correctly, with a partial explanation of differences from experimental results.	Sample space is incomplete or theoretical probabilities are incorrect, with no explanation of differences.
Use of the laws of probability	Chooses the addition law, complement rule and multiplication law correctly, justifies each by mutually exclusive or independent events, and calculates 0.7, 0.4, 0.3 and 0.49 correctly.	Applies the laws correctly with minor gaps in justifying why each one is allowed.	Mixes up the laws or applies them without checking the events, giving incorrect answers.
Tree diagram and two-day probabilities	Draws a fully labelled tree, multiplies along branches, checks that the total is 1.00, and finds 0.42 and 0.91 with the complement as a check.	Draws a correct tree and finds most of the outcomes correctly, with a missing check or small error.	Tree is incomplete or probabilities are not multiplied and added correctly.

Claim, Evidence, Reasoning and decisions	Gives a clear claim, uses specific evidence from the lessons, applies thresholds correctly for Wanjiru, Captain Otieno and Ms Achieng, and explains why decisions based on probability are sensible despite occasional bad luck.	Gives a claim with some evidence and mostly correct decisions, but the reasoning is brief.	Claim is unclear, evidence is missing or decisions are not linked to the probabilities.
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ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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